

tf.shape Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/shape

Returns a tensor containing the shape of the input tensor.

Function Signatures

```
tf.shape( input, out_type=None, name=None )
```

Code Examples

```
tf.shape(  
input, out_type=None, name=None  
)
```

```
tf.shape(  
input, out_type=None, name=None  
)
```

```
tf.shape(1.)
```

```
tf.shape(1.)
```

```
t = tf.constant([[[[1, 1, 1], [2, 2, 2]], [[3, 3, 3], [4, 4, 4]]]])  
tf.shape(t)
```

```
t = tf.constant([[[[1, 1, 1], [2, 2, 2]], [[3, 3, 3], [4, 4, 4]]]])
```

```
tf.shape(t)
```

Documentation

Returns a tensor containing the shape of the input tensor.

Used in the notebooks

- Effective Tensorflow 2
 - Ragged tensors
 - Introduction to Tensors
 - Customizing what happens in `fit()` with TensorFlow
 - Customizing what happens in `fit()`
 - DeepDream
 - Neural style transfer
 - Transfer learning with YAMNet for environmental sound classification
 - pix2pix: Image-to-image translation with a conditional GAN
 - Playing CartPole with the Actor-Critic method

See also `tf.size`, `tf.rank`.

`tf.shape` returns a 1-D integer tensor representing the shape of input.

For a scalar input, the tensor returned has a shape of `(0,)` and its value is the empty vector (i.e. `[]`).

For example:

In these cases, using `tf.Tensor.shape` will return more informative results.

(The first `None` represents the as yet unknown batch size.)

`tf.shape` and `Tensor.shape` should be identical in eager mode. Within `tf.function` or within a `compat.v1` context, not all dimensions may be known until execution time. Hence, when defining custom layers and models for graph mode, prefer the dynamic `tf.shape(x)` over the static `x.shape`.

Returns

